Anuroop Dasgupta

PhD Candidate in Astrophysics | ESO-Chile / Universidad Diego Portales

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RESEARCH SNAPSHOT

My research asks how disks of gas and dust turn into planetary systems, and what observable signatures trace that transformation. Three questions drive my work. When and where do planets first sculpt their disks? How do giant planets form, by slow core accretion or by rapid gravitational instability? And how do episodic accretion outbursts reshape the material planets are built from? To address these, I combine high-angular-resolution data across wavelengths, from ALMA in the millimeter to VLT/ERIS high-contrast imaging, VLTI/GRAVITY and CHARA interferometry, and JWST/MIRI spectroscopy. Together these trace disks from their outer dust reservoirs down to the innermost sub-au regions. I build the reduction and physical-modeling pipelines this work depends on, so that imaging, interferometry, spectroscopy, and time-domain surveys can all be brought to bear on the same question. My goal is to connect the observed structure of disks to the planets forming within them, and to carry these methods into the coming era of Rubin/LSST and space-based exoplanet imaging.

RESEARCH INTERESTS

- Protoplanetary disk structure and evolution: sizes, substructure, multiplicity, and evolutionary stage (ALMA).
- Substellar companions and early planet formation: high-contrast imaging and interferometric detection of embedded companions.
- Inner disk structure, accretion, and clearing on sub-au scales with optical/infrared interferometry (VLTI/GRAVITY, CHARA).
- Time-domain surveys and eruptive young stars: systematic searches for FU Orionis outbursts and their disks.

EDUCATION

March 2024 – Present

PhD in Astrophysics
European Southern Observatory & Universidad Diego Portales, Santiago, Chile

Structure and evolution of protoplanetary systems from infrared and millimeter observations, with an emphasis on high-angular-resolution interferometric studies of young stellar objects.

Supervisors: Dr. Lucas Cieza (UDP) & Dr. Aaron Labdon (ESO).

2021 – Nov 2023

M.Sc. in Physics
Presidency University, Kolkata, India

Specialization in astrophysics and cosmology. Thesis: “Metallicity Gradients in MaNGA Observed Dwarf Galaxies” (Supervisor: Dr. Ritaban Chatterjee).

2018 – Oct 2021

B.Sc. in Physics (Hons.)
Presidency University, Kolkata, India

Thesis: “The Thermal Evolution of the Intergalactic Medium in the Epoch of Reionization” (Supervisor: Dr. Saumyadip Samui).

PUBLICATIONS

First Author 2 published · 1 submitted · 1 to be submitted · 2 in preparation

- ∗**Dasgupta, A.**, Weber, P., Maio, F., et al. *Characterizing a New Companion Candidate Near V960 Mon. ApJL*, 2025. doi:10.3847/2041-8213/ade996 [ESO Press Release, July 2025]
- ∗**Dasgupta, A.**, Cieza, L. A., Bhowmik, T., et al. *Size Distributions of Protoplanetary Disks in the Ophiuchus Molecular Cloud. ApJL*, 2025. doi:10.3847/2041-8213/adb03c

Submitted / In Preparation

- ∗**Dasgupta, A.**, Labdon, A. *First K-band Interferometric Survey of Previously Unresolved Herbig Ae/Be Stars with VLTI/GRAVITY and CHARA.* To be submitted to A&A, 2026. [draft]
- **Dasgupta, A.**, Roy, N., Samui, S., Chatterjee, R., Bian, F. *Internal Structure, Not Environment, Drives Metallicity Gradients in Low-Mass Galaxies from MaNGA.* Submitted to A&A, 2026. [draft]
- **Dasgupta, A.**, et al. *Systematic Search for FU Orionis Outbursts in ZTF: A Multi-Step Photometric Pipeline and New Candidates.* In prep.
- **Dasgupta, A.**, Cieza, L., et al. *A JWST–ALMA View of the FUor HBC 687.* In prep.

Contributing Author 2 in press · 2 submitted · 1 published

- Bernardi, A., Zurlo, A., Lazzoni, C., Desidera, S., Mesa, D., Pérez, S., Nogueira, P. H., Barbato, D., **Dasgupta, A.** *SaNDi-SHoP: Searching for Satellites'N'Disks with a Star-Hopping Program II. Spectrophotometric Analysis and Orbital Monitoring of Directly Imaged Companions. A&A*, 2026. (in press).
- Lazzoni, C., Zurlo, A., Desidera, S., Bernardi, A., Pérez, S., Mesa, D., Barbato, D., Nogueira, P. H., **Dasgupta, A.** *SaNDi-SHoP: Searching for Satellites'N'Disks with a Star-Hopping Program I. A&A*, 2026. (in press). doi:10.48550/arXiv.2603.24796
- Bernardi, A., Zurlo, A., Cieza, L. A., Ruane, G., Christiaens, V., **Dasgupta, A.**, Guidi, G., Mawet, D., Pérez, S., Williams, J. P. *Searching for Embedded Protoplanets with the Keck/NIRC2 Vortex Coronagraph: Confirmation of a Planet in the Narrow Gap of the Elias 2-24 Disk.* Submitted to A&A, 2026.
- Bhowmik, T., Cieza, L. A., Miley, J. M., Nogueira, P. H., González-Ruilova, C., Chavan, P., Sierra, A., **Dasgupta, A.**, et al. *ODISEA: Substructures as a Function of SED Class and Disc Mass in 100 Systems.* Submitted to A&A, 2026.
- Orcajo, S., Cieza, L. A., Guilera, O. M., Pérez, S., González-Ruilova, C., Batalla-Falcon, G., Bhowmik, T., Chavan, P., Casassus, S., **Dasgupta, A.**, et al. *ODISEA: A Unified Evolutionary Sequence of Planet-Driven Substructures Explaining the Diversity of Disk Morphologies. ApJL*, 984, L57, 2025. doi:10.3847/2041-8213/adcd58

∗ Most significant publications.

CODE & SOFTWARE

2025 – Present

FU Orionis Discovery Pipeline — ZTF/ALeRCE
Python · end-to-end time-domain survey pipeline

Mines the full ZTF archive (∼131M sources via ALeRCE) for FU Orionis outbursts through a multi-stage selection cascade; recovers all known ZTF-era FUors and yields 245 vetted candidates, with a real-time monitor for ToO follow-up portable to Rubin/LSST.

2025 – Present

Temperature Gradient Disk Model — VLTI/CHARA Interferometry
Python · interferometric + SED modeling

Physically motivated disk model jointly fitting near-infrared visibilities and the broadband SED via a Hankel transform of the radial temperature profile, with Nelder–Mead and `emcee` MCMC, to constrain inner rim temperatures, radii, and gradients.

2024 – Present

ERIS/VLT L'-Band High-Contrast Imaging Pipeline
Python · ADI, PSF subtraction, sub-pixel astrometry

Angular differential imaging and PSF-subtraction pipeline with precision astrometry for VLT/ERIS, applied to V960 Mon to detect a new substellar companion candidate (ApJL 2025).

2025 – Present

JWST MIRI MRS Reduction & Analysis Pipeline
Python · IFU spectral-cube reduction and diagnostics

Custom reduction layer on the MINDS / `jwst` / VIP framework for MIRI MRS data of FUor stars, with extinction correction, continuum fitting, and dust, ice, and outflow diagnostics.

PRESS COVERAGE

July 2025

ESO Press Release eso2513 — "Astronomers witness newborn planet sculpting the dust around it"
European Southern Observatory | first-author result

National/international release featuring my first-author study of V960 Mon (ApJL 2025); release imagery credited to ESO/A. Dasgupta.

May 2025

ALMA Press Release — "ALMA inspires new models of how planet-forming disks evolve"
Atacama Large Millimeter/submillimeter Array | co-author result

Release on the ODISEA unified evolutionary sequence of planet-driven disk substructures (Orcajo et al., ApJL 2025), on which I am a co-author.

ACCEPTED TELESCOPE PROPOSALS

Principal Investigator

Period	Title
NTT/EFOSC2 (2026)	Spectroscopic Classification of New Young Eruptive Star Candidates
ESO P115 (2024)	VLT/VISIR: Formation and Distribution of Crystalline Material in Outbursting FUor Disks
ESO P114 (2024)	Satellites & Circumplanetary Disks Around DI Companions via Star-Hopping (Cont.)

Co-Investigator

Period	Title
ALMA Cyc. 11 (2025)	An ALMA and JWST/MIRI Line Survey of FUor Objects: The Chemistry Revealed by Stellar Outbursts (2025.1.00299.S)
ESO P117 (2026)	ERIS/VLT: A Direct Test of Models Linking Disc Substructures to Planet Formation through Core Accretion
ESO P116 (2026)	GRAVITY/VLTI: Resolving the Inner Disks of a Complete Sample of Planet-Forming Disks
ESO P115 (2024)	Searching for PDS 70 Analogues
ESO P114 (2024)	First Survey of (Sub-)Stellar Companions Around Protostars (Part II)

CONFERENCES, TALKS & COLLOQUIA

Colloquia & Seminars: 4 · Talks & Posters: 9 (7 international, 2 national)

Jul 2026

Discs on the Exe, University of Exeter
Exeter, UK | *V960 Mon: Ideal Laboratory for Gravitational Instability (talk); First Volume-Complete Survey on Nearby Herbigs with GRAVITY (poster)*

May 2026

Eruptive Young Stars Workshop, Konkoly Observatory
Budapest, Hungary | *JWST–ALMA View of HBC 687 (contributed talk)*

Feb 2026

Spirit of Lyot 6, Caltech
Pasadena, CA, USA | *V960 Mon: Ideal Laboratory for Gravitational Instability (poster)*

Nov 2025

Stellar Variability: Taking the Pulse of the Universe, IUCAA
Pune, India | *V960 Mon: Ideal Laboratory for Gravitational Instability (contributed talk)*

Sep 2025

Indian Institute of Technology Bombay
Mumbai, India | *Planet Formation Across Different Wavelengths (invited colloquium)*

Sep 2025

UKI Discs Meeting, University of Hertfordshire
Hertfordshire, UK | *Complete Size Distribution for the 100 Brightest Protoplanetary Disks in Ophiuchus (contributed talk)*

Oct 2025

IISER Kolkata
Kolkata, India | *Planet Formation Across Different Wavelengths (invited colloquium)*

Feb 2025

Circumplanetary Disks and Satellite Formation III
Kyoto, Japan | *Complete Size Distribution for the 100 Brightest Protoplanetary Disks in Ophiuchus (contributed talk)*

Dec 2024

Tata Institute of Fundamental Research (TIFR)
Mumbai, India | *Size Distributions of Protoplanetary Disks in Nearby Star-Forming Regions (invited colloquium)*

Dec 2024

National Centre for Radio Astrophysics (NCRA)
Pune, India | *Size Distributions of Protoplanetary Disks in Nearby Star-Forming Regions (invited colloquium)*

Nov 2024

SOCHIAS XIX Meeting, Universidad de Tarapacá
Arica, Chile | *What is the Size Distribution of Disks in Nearby Star-Forming Regions? (contributed talk)*

Nov 2024

Unveiling the Origin of Brown Dwarfs, ESO
Santiago, Chile | *First Direct Observational Evidence of Gravitational Instability? (contributed talk)*

OBSERVING & INSTRUMENTATION

March 2026

Night Astronomer — NTT/EFOSC2, La Silla Observatory
ESO-Chile, La Silla, Chile

Conducted service observations over 10 nights, executing 13 approved ESO-Chile programs spanning optical imaging and low/medium-resolution spectroscopy, including target acquisition, real-time instrument operation, and quality control.

2025 – Present

GRAVITY Acquisition-Camera Photometry — VLTI/GRAVITY
ESO & UDP | instrument development

Developing reliable photometric calibration from the GRAVITY acquisition camera in dual-field mode, identifying NIR standards suitable for dual-field observations, preparing and executing test OBs, and assessing flux-calibration stability and accuracy, toward formalising acquisition-camera photometry as a standard science capability.

July 2025

Night Astronomer — VLTI/GRAVITY, Paranal Observatory
ESO, Atacama Desert, Chile

Served as night astronomer over 8 nights, taking part in VLTI operations and GRAVITY instrument calibration, including on-sky calibration sequences and real-time instrument operation.

Oct 2024

Swope Telescope Observing Run
Las Campanas Observatory, Chile

Five nights of multi-filter imaging for a project on the metallicity–dynamics connection in galaxies.

SUPERVISION

2025 – Present

MSc Thesis Co-Supervisor — Srimoyree Mitra (St. Xavier's College, Kolkata, India)
Universidad Diego Portales

"Variability Studies of Class I Young Stellar Objects in the Orion Nebular Cluster Region." Multi-epoch photometric variability analysis across disk evolutionary stages.

TECHNICAL SKILLS

Facilities ALMA (Bands 6–8), VLT/ERIS, VLTI/GRAVITY, CHARA/MIRCX-MYSTIC, VLT/VISIR, JWST/MIRI, NTT/EFOSC2, Swope

Programming Python (primary), Fortran, Mathematica, LaTeX, Bash; Git, test-driven development

Astro tools CASA, PMOIRED, `jwst`/MINDS/VIP, Astropy, Specutils, `emcee`, IRAF, DS9, TOPCAT, Streamlit

Analysis Interferometric visibility modeling · Bayesian inference / MCMC · Lomb–Scargle · multiwavelength photometry (ZTF, WISE, 2MASS, Gaia, PanSTARRS) · spectral line fitting · bootstrap / Monte Carlo

Pipelines End-to-end instrument-adaptable workflows; large-scale DB queries (SQL, ALeRCE API); automated real-time alert systems

AWARDS, FELLOWSHIPS & HONOURS

- **ANID Beca de Doctorado Nacional** (National Doctoral Fellowship) — Agencia Nacional de Investigación y Desarrollo (ANID), Chile, 2025.
- **ESO Studentship Programme** — European Southern Observatory, 2025–2027.
- **Outstanding Paper Award** — awarded for the best paper/presentation at the state level (West Bengal, India); 31st West Bengal State Science & Technology Congress, 2024.
- **Paramesh Chandra Bhattacharya Memorial Award** — Best Master's thesis, Presidency University, 2023.
- **Academic Ranking** — ranked 3rd in M.Sc. semesters 3 & 4, Presidency University; ranked 14th/~1,000 in Pune University MSc entrance (declined).
- **Top 1% West Bengal State Board**, Class 10 & 12 (2018); selected for INSPIRE *Young Scientists Meet*, DST India, 2016.

OUTREACH

- Great Asteroid Festival (5th ed.) — AUI/NRAO/YEMS outreach stand, Las Vizcachas, Puente Alto, Chile (2026).
- Colloquium Host, ESO-Chile weekly colloquium programme (ongoing).
- "Exploring Exoplanets" (PME-funded, YEMS Millennium Nucleus "Astrophysics at School") — developing an educational card game and activity book on exoplanets, disks, and Solar System moons for underserved communities (ongoing).
- Public telescope observations and guided eclipse viewing (Universidad Diego Portales); science communication and online seminars introducing astrophysics to students in India and Chile.

SCHOOLS & WORKSHOPS

Summer School in Astrophysics — Indian Institute of Astrophysics, Bangalore *2022*

Summer Research Fellowship (High Energy Astrophysics) — IISER Bhopal *2022*

International Capsule Workshop on Astrobiology — Indian Astrobiology Research Centre *2019*

REFERENCES

Dr. Lucas Cieza
Professor & PhD Supervisor
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Dr. Aaron Labdon
Staff Astronomer & ESO Studentship Supervisor
European Southern Observatory
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Dr. Alice Zurlo
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Assistant Professor
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